Continuity & Differentiability: Formula Sheet

Continuity

Continuous at a: $f(a^-) = f(a^+) = f(a)$

Left continuous: $f(a^-) = f(a)$

Right continuous: $f(a^+) = f(a)$

In $[a, b]$: cont. in (a, b), $f(a^+) = f(a)$, $f(b^-) = f(b)$

Checkpoints

$[f(x)], \{f(x)\}$: $f(x) \in I$

$\operatorname{sgn} f(x)$: $f(x) = 0$

$|f(x)|$ (derivability): $f(x) = 0$

$f(g(x))$: g discont., $g(x) =$ discont. of f

Built-in: $x^{2n} \to 0, 1, \infty$ for $|x| < 1, = 1, > 1$

Discontinuity

Removable: $\lim f$ exists, $\neq f(a)$ or $f(a)$ undefined

Missing point: $f(a)$ undefined

Isolated point: $f(a) \neq \lim f$

Jump: $|f(a^+) - f(a^-)|$

Infinite: $f(a^\pm) = \pm\infty$; oscillatory: $\sin \frac{1}{x}$

Theorems

EVT: cont. on $[a, b] \Rightarrow m \leq f \leq M$ attained

IMVT:

$$f(a)f(b) < 0 \Rightarrow \exists c \in (a, b),\ f(c) = 0$$

Fixed point: $g(x) = f(x) - x$

Derivability

RHD:

$$\lim_{h \to 0^+} \frac{f(a+h) - f(a)}{h}$$

LHD:

$$\lim_{h \to 0^+} \frac{f(a-h) - f(a)}{-h}$$

Other form:

$$\lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

$$\lim_{h \to 0} \frac{f(a+ph) - f(a+qh)}{h} = (p-q)f'(a)$$

Derivable: LHD = RHD = finite

Quick concept: $f'(a^\pm) = \lim_{x \to a^\pm} f'(x)$

LHD, RHD and continuity

Equal, finite: derivable

Finite, unequal: continuous, sharp corner

One infinite: not derivable, check continuity

Derivable $\Rightarrow$ continuous

Standard results

$x^p \sin \frac{1}{x}$: cont. $p > 0$, deriv. $p > 1$, f' cont. $p > 2$

$g|f|$, $f(a) = 0$: derivable iff $g(a) = 0$

Q/Q': cont. where $g = h$; deriv. where also $g' = h'$

$|x|$: corner at 0; $x^{1/3}$: vertical tangent

$\sum |x - a_i|$: non-derivable at each a_i

Functional equations

$f(x+y) = f(x) + f(y)$: $f = f'(0)x$

$f(x+y) = f(x)f(y)$: $f = e^{f'(0)x}$

$f(xy) = f(x) + f(y)$: $f = f'(1) \ln x$

$f(xy) = f(x)f(y)$: $f = x^{f'(1)}$

$f(x+y) = f(x) + f(y) + 2xy - 1$, $f'(0) = 4$: $x^2 + 4x + 1$